

EXPLORATORY DATA ANALYSIS

E-Commerce Sales Dataset

Project 2 – Data Analytics

DecodeLabs – Data Analytics Training Program

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Python

Pandas

NumPy

Matplotlib

Seaborn

1. Introduction

Data plays a critical role in helping businesses make informed decisions. However, raw data often contains hidden patterns and valuable insights that cannot be understood without proper analysis.

Exploratory Data Analysis (EDA) is the process of examining datasets to summarize their main characteristics, identify trends, detect anomalies, and discover relationships between variables through statistical techniques and visualizations.

This project focuses on analyzing an E-Commerce Sales Dataset containing customer orders, product details, payment methods, and transaction information. The objective is to understand business performance, customer purchasing behavior, and revenue trends.

2. Objective

The primary objectives of this project are:

- To perform Exploratory Data Analysis (EDA) on the provided dataset.
- To understand the structure and quality of the dataset.
- To calculate descriptive statistics such as mean, median, and count.
- To identify trends and patterns in sales data.
- To detect outliers and unusual transactions.
- To analyze product performance and revenue generation.
- To generate business insights through visualizations.

3. Dataset Description

The dataset contains e-commerce transaction records collected over a specific period.

Dataset Information

Attribute	Description
Rows	1200
Columns	14
Dataset Type	E-Commerce Sales Data
Format	Excel (.xlsx)

Key Columns

- Order ID
- Customer ID
- Date
- Product

- Quantity
- Unit Price
- Total Price
- Payment Method
- Order Status
- Coupon Code
- Referral Source

The dataset represents customer purchases across different products and payment methods.

4. Tools and Technologies Used

Category	Details
Programming Language	Python
Libraries	Pandas, NumPy, Matplotlib, Seaborn
Development Environment	Jupyter Notebook

5. Data Inspection and Quality Assessment

The first step involved understanding the structure of the dataset using:

- `df.head()`
- `df.info()`
- `df.describe()`

Missing Values Analysis

The dataset was checked for missing values.

- ✓ *Missing values were identified mainly in the CouponCode column.*
- ✓ *These missing values indicate that many customers did not use promotional coupons while placing orders.*

Duplicate Records Analysis

- ✓ *No significant duplicate records were found.*
- ✓ *The dataset maintained good data quality.*

6. Descriptive Statistical Analysis

Descriptive statistics provide a summary of numerical data. The following measures were calculated:

Measure	Description
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Mean	Average value across transactions
Median	Middle value in ordered dataset
Minimum	Lowest recorded value
Maximum	Highest recorded value
Standard Deviation	Spread of values around the mean
Count	Total number of valid records

- ✓ Average quantity purchased per order was approximately 3 units.
- ✓ The average order value exceeded Rs.1000, indicating healthy transaction volumes.
- ✓ Most customers purchased between 1 and 5 items per order, with moderate variation.

7. Product Analysis

Visualization Used

- Bar Chart
- Pie Chart

Findings

- ✓ Printer was identified as the most frequently ordered product.
- ✓ Tablet and Chair also recorded strong sales volumes.
- ✓ Product demand varied across categories.

Business Insight: Understanding product popularity helps businesses optimize inventory management and marketing strategies.

8. Payment Method Analysis

Visualization Used

- Bar Chart
- Pie Chart

Findings

- ✓ Online payment methods were more commonly used than traditional options.
- ✓ Customers preferred convenient digital transactions.

Business Insight: The company should continue improving digital payment experiences to enhance customer satisfaction.

9. Order Status Analysis

Visualization Used

- Bar Chart

Findings

- ✓ *Categories analyzed: Delivered, Shipped, Pending, Returned, Cancelled.*
- ✓ *The distribution of order statuses was relatively balanced.*

Business Insight: Returned and cancelled orders should be monitored carefully to reduce operational losses.

10. Sales Trend Analysis

Visualization Used

- Line Chart
- Area Chart

Findings

- ✓ *Revenue showed a positive growth trend over time.*
- ✓ *Sales increased significantly during the later months of the dataset.*
- ✓ *June 2025 recorded the highest revenue among recent periods.*

Business Insight: Increasing sales indicate business growth and expanding customer demand.

11. Distribution Analysis

Visualization Used

- Histogram

Findings

- ✓ *Most transactions occurred within lower and medium price ranges.*
- ✓ *A smaller number of transactions had significantly higher values.*

Business Insight: The business primarily serves regular customers while also attracting some high-value purchases.

12. Outlier Detection

Visualization Used

- Box Plot

Findings

- ✓ *Several transactions had unusually high order values.*
- ✓ *These transactions appeared significantly above the normal transaction range.*

Business Insight: High-value orders may represent premium customers or bulk purchases and should be studied separately.

13. Correlation Analysis

Visualization Used

- Correlation Heatmap

Findings

- ✓ *Unit Price showed a strong positive relationship with Total Price.*
- ✓ *Quantity also demonstrated a positive correlation with Total Price.*

Business Insight: Revenue is largely influenced by product pricing and quantity purchased.

14. Revenue Analysis by Product

Visualization Used

- Bar Chart
- Horizontal Bar Chart

Findings

- ✓ *Some products generated high revenue despite having fewer orders.*
- ✓ *Premium-priced products contributed significantly to overall revenue.*

Business Insight: Product profitability should be considered alongside sales volume for strategic decision-making.

15. Key Findings

1. The dataset contains 1200 e-commerce orders.
2. Product demand varies significantly across categories.
3. Printer emerged as the most frequently ordered product.
4. Online payment methods were the preferred mode of payment.
5. Sales exhibited a positive growth trend over time.
6. High-value transactions were identified as outliers.
7. Quantity and Unit Price strongly influence Total Revenue.
8. Most transactions occur within moderate price ranges.
9. Revenue generation differs across product categories.
10. Data quality was generally good with minimal issues.

16. Conclusion

This project successfully applied Exploratory Data Analysis (EDA) techniques to an E-Commerce Sales Dataset. Through data inspection, descriptive statistics, trend analysis, outlier detection, and visualization, meaningful business insights were uncovered.

The analysis revealed customer purchasing patterns, product performance trends, payment preferences, and revenue-driving factors. The findings demonstrate how data analytics can support informed business decisions and improve operational efficiency.

Overall, this project strengthened practical skills in Python, data analysis, visualization, statistical interpretation, and business insight generation, providing a strong foundation for future data analytics projects.

17. Skills Demonstrated

Python Programming	Pandas
NumPy	Data Cleaning
Data Analysis	Exploratory Data Analysis (EDA)
Data Visualization	Statistical Analysis
Business Intelligence	Problem Solving

Thank You